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CLASS : CSAI-D

ROLL : 34

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```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score


dataset = pd.read_csv("temperatures.csv")

print("First 5 rows:\n", dataset.head())


X = dataset[['JAN','FEB','MAR','APR','MAY','JUN',
             'JUL','AUG','SEP','OCT','NOV','DEC']]


y = dataset['ANNUAL']


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


model = LinearRegression()
model.fit(X_train, y_train)


y_pred = model.predict(X_test)


mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2  = r2_score(y_test, y_pred)

print("\nModel Performance:")
print("MSE :", mse)
print("MAE :", mae)
print("R2 Score :", r2)



# Actual vs Predicted
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Temperature")
plt.ylabel("Predicted Temperature")
plt.title("Actual vs Predicted Temperature")


z = np.polyfit(y_test, y_pred, 1)
p = np.poly1d(z)
plt.plot(y_test, p(y_test))

plt.show()
```

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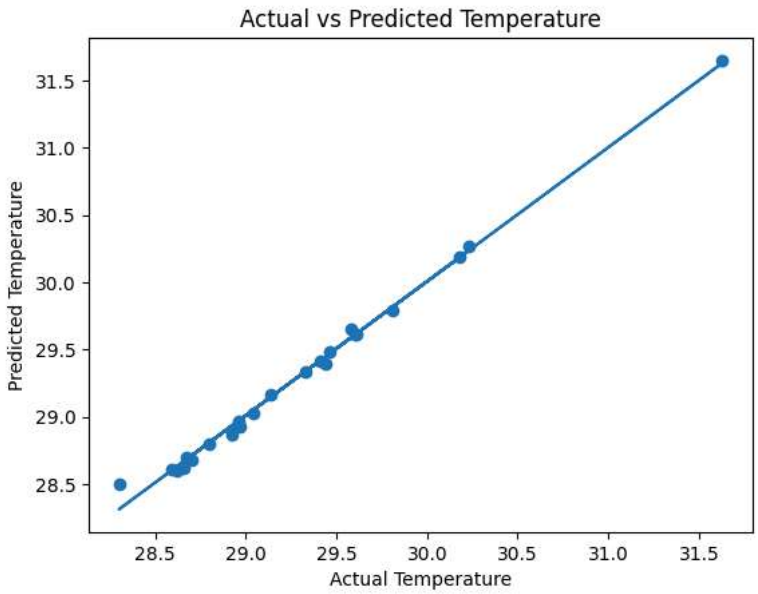
```
First 5 rows:
    YEAR    JAN    FEB    MAR    APR    MAY    JUN    JUL    AUG    SEP    OCT  \
0  1901  22.40  24.14  29.07  31.91  33.41  33.18  31.21  30.39  30.47  29.97
1  1902  24.93  26.58  29.77  31.78  33.73  32.91  30.92  30.73  29.80  29.12
2  1903  23.44  25.03  27.83  31.39  32.91  33.00  31.34  29.98  29.85  29.04
3  1904  22.50  24.73  28.21  32.02  32.64  32.07  30.36  30.09  30.04  29.20
4  1905  22.00  22.83  26.68  30.01  33.32  33.25  31.44  30.68  30.12  30.67

     NOV    DEC  ANNUAL  JAN-FEB  MAR-MAY  JUN-SEP  OCT-DEC
0  27.31  24.49   28.96    23.27    31.46    31.27    27.25
1  26.31  24.04   29.22    25.75    31.76    31.09    26.49
2  26.08  23.65   28.47    24.24    30.71    30.92    26.26
3  26.36  23.63   28.49    23.62    30.95    30.66    26.40
4  27.52  23.82   28.30    22.25    30.00    31.33    26.57

Model Performance:
MSE : 0.002440835094691285
MAE : 0.029590158623251355
R2 Score : 0.9949515631120867
```




Start coding or generate with AI.

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